Lab 6:Neutron (Networking)

Neutron:

Neutron is an OpenStack project to provide “Network connectivity as a Service (NaaS)”. The OpenStack Networking service (neutron) provides an API that allows users to build rich networking topologies, set up and define network connectivity, configure advanced network policies and addressing in the cloud.

OpenStack Networking handles the creation and management of a virtual networking infrastructure, including networks, switches, subnets, and routers for devices managed by the OpenStack Compute service (nova). Advanced services such as firewalls or virtual private network (VPN) can also be used but are not currently available on this cloud.

OpenStack Networking integrates with various OpenStack components:

- OpenStack Identity service (keystone) is used for authentication and authorization of API requests.
- OpenStack Compute service (nova) is used to plug each virtual NIC on the VM into a particular network.
- OpenStack Dashboard (horizon) is used by administrators and project users to create and manage network services through a web-based graphical interface.

Switch:

A switch is a device in a computer network that connects other devices together. Multipledata cables are plugged into a switch to enable communication between different networked devices. Switches manage the flow of data across a network by transmitting a receivednetwork packet only to the one or more devices for which the packet is intended. Each networked device connected to a switch can be identified by its network address, allowing the switch to direct the flow of traffic maximizing the security and efficiency of the network.

A switch is more intelligent than an Ethernet hub, which simply retransmits packets out of every port of the hub except the port on which the packet was received, unable to distinguish different recipients, and achieving an overall lower network efficiency. An Ethernet switch operates at the data link layer (layer 2) of the OSI model to create a separate collision domain for each switch port. A device that also operates at these higher layers is known as a multilayer switch.

Router:

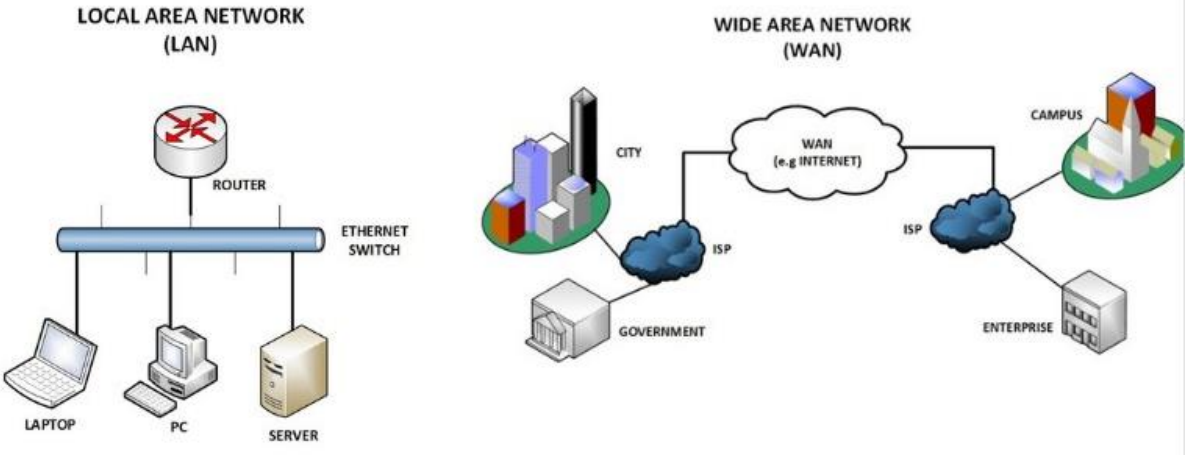
A router is a networking device that forwards data packets between computer networks. Routers perform the traffic directing functions on the Internet. Data sent through the internet, such as a web page or email, is in the form of data packets. A packet is typically forwarded from one router to another router through the networks that constitute an internetwork until it reaches its destination node.

A router is connected to two or more data lines from different networks. When a data packet comes in on one of the lines, the router reads the network address information in the packet to determine the ultimate destination. Then, using information in its routing table or routing policy, it directs the packet to the next network on its journey.

Network:

A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

Two very common types of networks include:



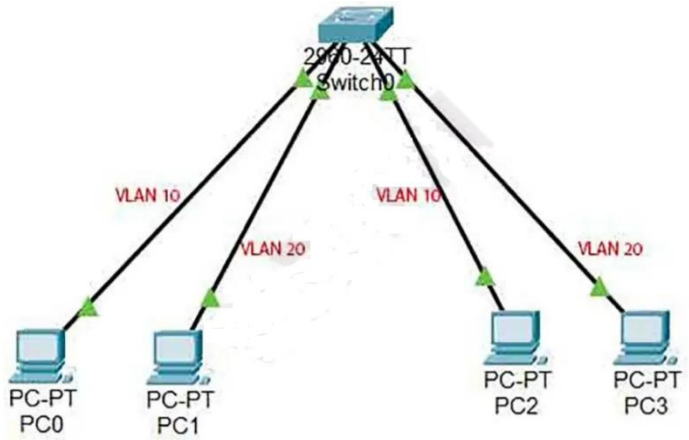
Vlans:

A VLAN is a logical grouping of network users and resources connected to administratively defined ports on a switch.

The number of broadcast domains are increased with VLANs while the size decreases.

In the VLAN a group of users with the demand of high security can be included so that the external users out the VLAN cannot interact with them.

Every VLAN must have different network.



IP addressing:

IP address is an add t, especially outside the LAN. An IP address is a 32 bit unique address.

Types of IP Addressing

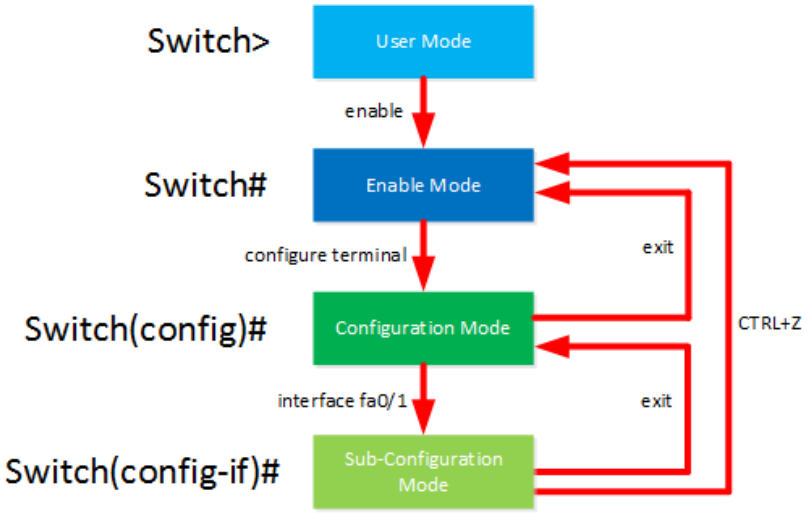
Private: A private IP address is the address space allocated by InterNIC to allow organizations to create their own private network. There are three IP blocks (1 class A, 1 class B and 1 class C) reserved for a private use.

Summary of IPv4 Classes

	Public IP Range	Private IP Range	Subnet Mask	# of Networks	# of Hosts per Network
Class A	1.0.0.0 to 127.0.0.0	10.0.0.0 to 10.255.255.255	255.0.0.0	126	16,777,214
Class B	128.0.0.0 to 191.255.0.0	172.16.0.0 to 172.31.255.255	255.255.0.0	16,382	65,534
Class C	192.0.0.0 to 223.255.255.0	192.168.0.0 to 192.168.255.255	255.255.255.0	2,097,150	254

Public: A public IP address is the address that is assigned to a computing device to allow direct access over the Internet. A web server, email server and any server device directly accessible from the Internet are candidate for a public IP address. A public IP address is globally unique and can only be assigned to a unique device.

Switch Configuration Modes:



How to enabled the switch?

Switch>en

Switch# show run

How to create the Vlan?

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 30
```

How to create the instance of Vlan?

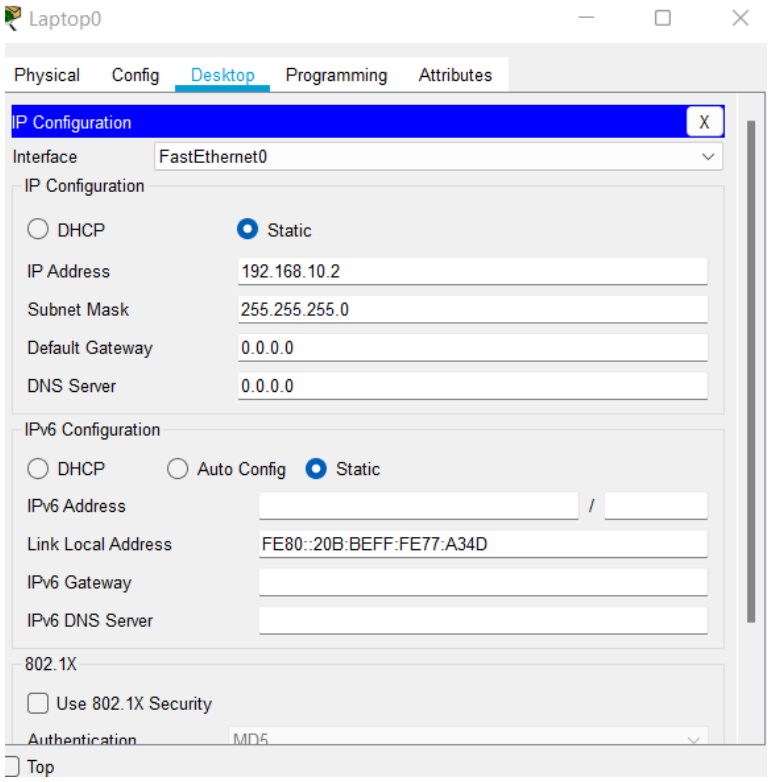
```
Switch(config-vlan)#interface fastEthernet 0/3
Switch(config-if)#switch mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

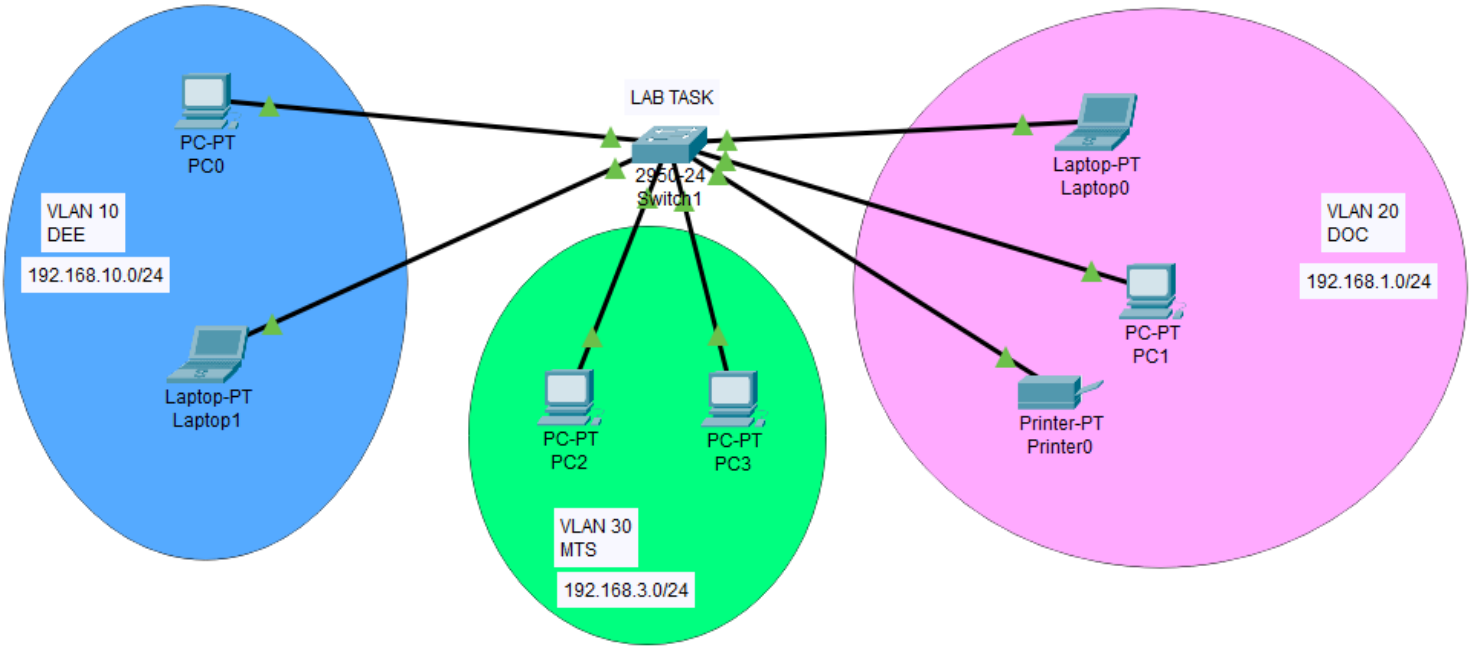
```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/4, Fa0/5,
Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9,
Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13,
Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17,
Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21,
Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1,
Gig0/2
10   students                         active    Fa0/1
20   faculty                          active    Fa0/2
30   VLAN0030                         active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Set the IP address on devices

Click on device-> desktop -> IP configuration



Task:

- Create 3 different VLANS as shown in the figure.
- Assign ip's to end systems according to network id's given.
- Check the connectivity of different vlans, and ping.